

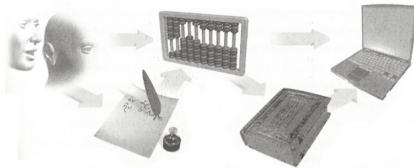
Natural Computing

LMU Munich
summer term 2025

Thomas Gabor



Artificial Chemistries



“The history of the universe can be thought of as a sequence of information processing revolutions, each of which builds on the technology of the previous ones.”

Seth Lloyd. *Programming the Universe*. Vintage Books, 2006.

On Multi-Sets...

Let X be a set with members from a space $\mathcal{X}$.

We can identify a set X with its membership function $\mu_X : \mathcal{X} \rightarrow \mathbb{B}$ such that

$$\mu_X(x) = 1 \iff x \in X.$$

For a multiset X' we generalize the membership function so that

$$\mu_{X'}(x) \geq 1 \iff x \in X'.$$

In both cases, the subset relation between (multi-)sets X, Y can be defined as

$$X \subseteq Y \iff \forall x \in X : \mu_X(x) \leq \mu_Y(x).$$

We then define $\wp(X) = 2^X = \mathbb{B}^X \cong X \rightarrow \mathbb{B}$,

and $\wp^*(X) = \mathbb{N}^X \cong X \rightarrow \mathbb{N}$,

to be the powerset functions for sets and multisets, respectively.

More on Multi-Sets...

Definition 10 (artificial chemistry, soup). Let $\mathcal{X}$ be a state space. Let $\mathcal{R}$ be a set of reaction rules $R \in \mathcal{R}$ where each R is a function $R : \mathcal{X}_R^{\text{dom}} \rightarrow \mathcal{X}_R^{\text{cod}}$ for $\mathcal{X}_R^{\text{dom}}, \mathcal{X}_R^{\text{cod}} \subseteq \wp^*(\mathcal{X})$. Let $\mathcal{A}(X, \mathcal{R})$ be the set of all *reaction materials* from $X \in \wp^*(\mathcal{X})$ for all rules in $\mathcal{R}$, i.e.,

$$\mathcal{A}(X, \mathcal{R}) = \{(X', R) \mid X' \subseteq X \wedge X' \in \mathcal{X}_R^{\text{dom}} \wedge R \in \mathcal{R}\}.$$

Let $A : \wp^*(\mathcal{X}) \rightarrow \wp^*(\mathcal{X}) \times \mathcal{R}$ be a possibly randomized or non-deterministic function that takes a multi-set $X \in \wp^*(\mathcal{X})$ and returns a multi-set of reactants and a reaction rule to perform so that $A(X) \in \mathcal{A}(X, \mathcal{R})$. The tuple $(\mathcal{X}, \mathcal{R}, A)$ defines an artificial chemistry. A multi-set $X \in \wp^*(\mathcal{X})$ of an artificial chemistry is also called soup; each element $x \in X$ is also called a particle. The evolution of a soup for g generations is a sequence $\langle X_0, \dots, X_g \rangle$ so that X_0 is given as the initial soup and, for all t with $0 \leq t \leq g - 1$

$$X_{t+1} = (X_t \setminus X') \uplus R(X')$$

where $(X', R) = A(X_t)$.

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where $(X', R) = A(X_t)$.

Reaction rules are usually written in the form $R = \sum_i x_i \rightarrow \sum_j y_j$ for $x_i, y_j \in \mathcal{X}$, $i, j \in \mathbb{N}$, $\{x_i\}_i \in \mathcal{X}_R^{\text{dom}}$, $\{y_j\}_j \in \mathcal{X}_R^{\text{cod}}$, and $R(\{x_i\}_i) = \{y_j\}_j$.

Example 1 ([2]). We imagine a colony of chameleons who can either assume a red, green, or blue color. Whenever two chameleons with a different color meet, they assume the third color (that none of them has).

Let $\mathcal{X} = \{r, g, b\}$.

Let $\mathcal{R} = \{R_1, R_2, R_3, R_4, R_5, R_6\}$ with

$$R_1 = r + r \rightarrow r + r,$$

$$R_2 = r + g \rightarrow b + b,$$

$$R_3 = r + b \rightarrow g + g,$$

$$R_4 = b + b \rightarrow b + b,$$

$$R_5 = b + g \rightarrow r + r,$$

$$\text{and } R_6 = g + g \rightarrow g + g.$$

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

Let $X_0 \sim \wp^*(\mathcal{X})$.

Experiment Time!

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early results

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late results

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Experiment Time! *again*

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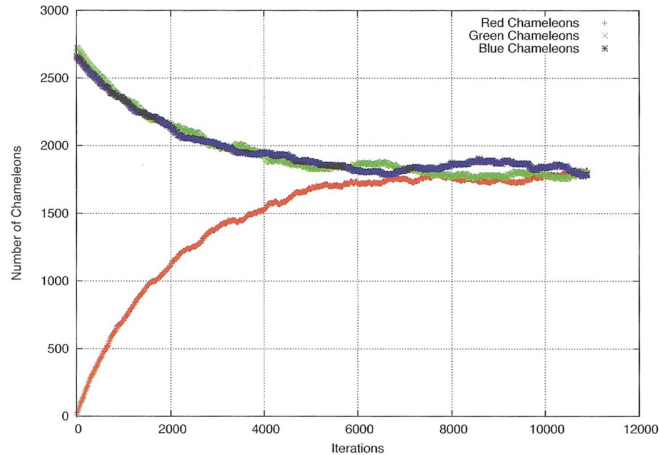


Figure 2.5 The first two generations (equivalent to 10,800 iterations) of an explicit simulation of encounters between chameleons changing colors. A total of 5,400 chameleons is simulated, starting from a condition where only blue and read chameleons are present, in virtually equal numbers, due to randomized initial condition. Red chameleons quickly pick up in numbers, until only fluctuations are left between numbers.

Example 2 ([2]). A soup of natural numbers divides each other...?

Let $\mathcal{X} = \mathbb{N} \setminus \{0, 1\}$.

Let $\mathcal{R} = \{x_1 + x_2 \rightharpoonup x_1 + x_3 : x_1, x_2, x_3 \in \mathcal{X} \wedge x_1 < x_2 \wedge x_3 = x_2/x_1\}$

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

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results?

Example 3. A soup of single-digit numbers multiplies each other...?

Let $\mathcal{X} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Let $\mathcal{R} = \{x_1 + x_2 \rightarrow x_3 + x_3 : x_1, x_2, x_3 \in \mathcal{X} \wedge x_3 = x_1 \cdot x_2 \bmod 10\}$

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

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Stirring Soup

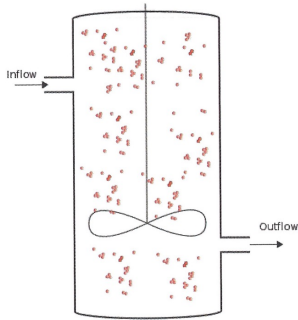


Figure 7.3 Well-stirred flow reactor with dilution — inflow and outflow of material.

Stochasticity

No-Ops, Creation, and Destruction

Example 4. A soup of functions writing into each other...?

Let $\mathcal{X} = \mathcal{X} \rightarrow \mathcal{X}$, i.e., the space of all functions that can take functions and return new functions.

Let $\mathcal{R} = \{x_1 + x_2 \rightarrow x_1 + x_3 \mid x_1, x_2, x_3 \in \mathcal{X} \wedge x_3 = x_1(x_2)\}$, i.e., when two functions x_1, x_2 meet, x_2 is overwritten with the result of applying x_1 to x_2 .

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

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Example: λ -calculus

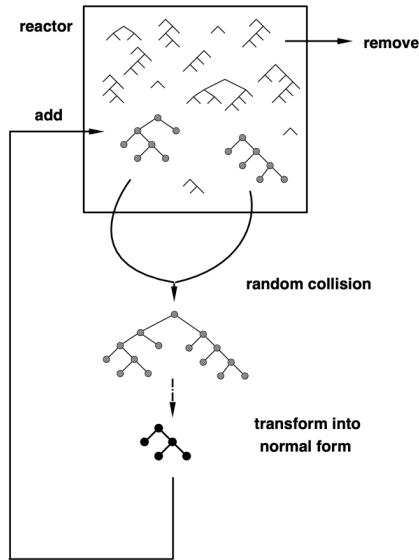


Figure 1: λ -calculus flow reactor. Two expressions A and B are chosen at random and a new object, $(A)B$, is constructed by “application” (see appendix A.2). Putting $(A)B$ into its normal form by β -reduction (see appendix A.2), effectively decides which object species (i.e., “stable molecular formula”) the new object is an instance of.

On λ -calculus...

Example:

$$(\lambda x. f \ x) \ y \quad \longrightarrow \quad f \ y$$

Example in Python:

$$(\text{lambda } x: \ f(x))(y) \quad \longrightarrow \quad f(y)$$

Example: λ -calculus

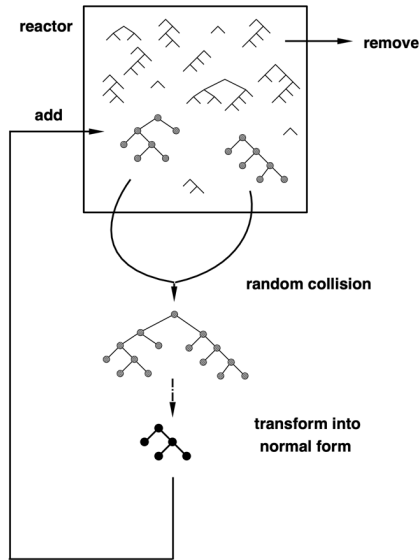


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Experiment Time??

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Stability

Mere Stability vs. Self-Replication

Example: λ -calculus

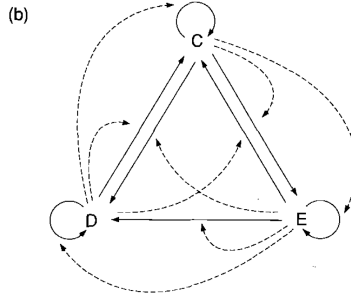
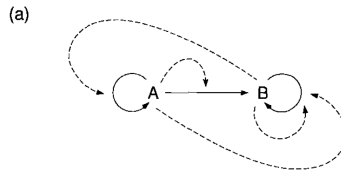


Figure 1. Level 0 ecology. A solid arrow denotes a transformation from argument object to value object. A dotted arrow connects the operator object with the transformation performed by it. Thus, $E \circ C = D$ is shown as a solid arrow from C to D pointed at by a dotted arrow originating in E.

$$\begin{aligned}
 A &\equiv \lambda x_1 . (x_1) \lambda x_2 . \lambda x_3 . (x_3) \lambda x_4 . \lambda x_5 . (x_5) x_4, & B &\equiv \lambda x_1 . (x_1) \lambda x_2 . \lambda x_3 . (x_3) x_2, \\
 C &\equiv \lambda x_1 . \lambda x_2 . (x_1) \lambda x_3 . x_1, & D &\equiv \lambda x_1 . \lambda x_2 . \lambda x_3 . (x_2) \lambda x_4 . x_2, \\
 E &\equiv \lambda x_1 . \lambda x_2 . \lambda x_3 . \lambda x_4 . (x_3) \lambda x_5 . x_3.
 \end{aligned}$$

Properties of Self-Replicators

longevity

fecundity

copy-fidelity

Levels of Organization

Level 0: reproduction and ecology

- cycle of replicators (easily collapses to only one replicator)

Level 1: self-maintenance and organization

- emerges when mere copy actions are forbidden
- structure and interaction between particles can be described with specific rules
- part of the organization can generate all its elements anew

Level 2: organization of organizations

- create L1 organizations independently and then put them into the same soup
- organizations maintain themselves separately but interact via “glue” particles

Emergence of Implicit Target Functions

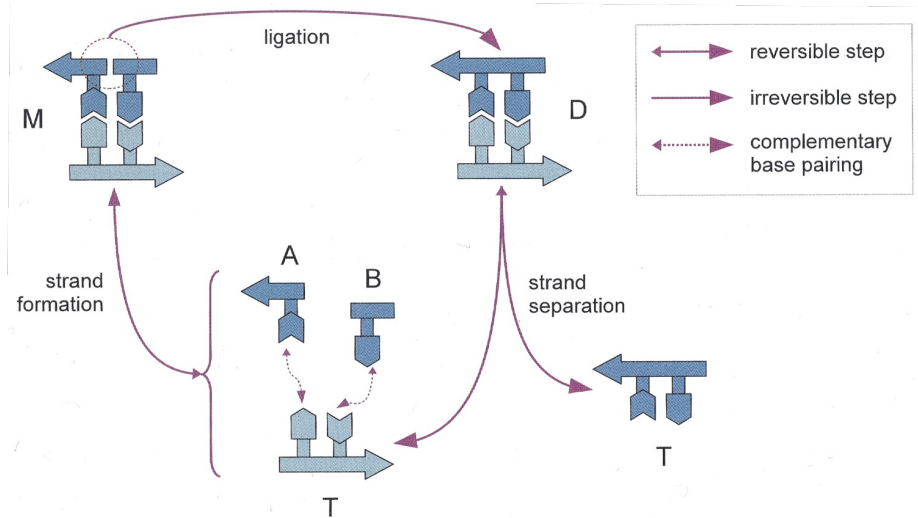


Figure 6.6 A model of the minimal self-replicator demonstrated by Von Kiedrowski [890]: The two *T* molecules are complementary templates (hexamers in the experiments). Each of them base-pairs with units *A* and *B* (trimers in the experiments), to form an intermediate molecule *M* which is almost a double strand: the still missing link between *A* and *B* is then catalyzed by *T* itself, forming the double strand *D*. *D* then splits into two single strands *T* and the cycle repeats. Adapted from [890].

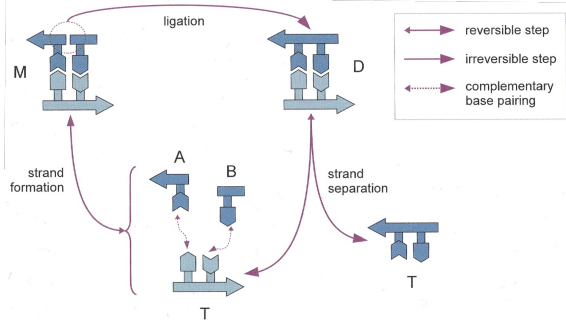


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Example 5. A DNA soup.

Let $\mathcal{X} = \{A, C, G, T\}^{2 \times 2} \cup \{A, C, G, T\}^2 \cup \{A, C, G, T\}^1$.

Let $\mathcal{R} = \{R_{1,-,-}, R_{2,-,-}, R_{3,-,-}, R_{4,-,-}\}$ with

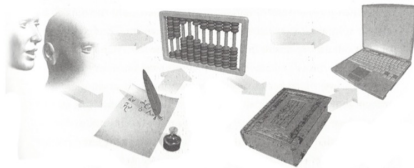
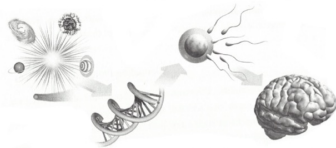
$$\begin{aligned}
 R_{1,X,Y} &= (X \ Y) + (-X) + (-Y) && \xrightarrow{0.5} \begin{pmatrix} X & Y \\ -X & -Y \end{pmatrix} \\
 R_{2,X,Y} &= \begin{pmatrix} X & Y \\ -X & -Y \end{pmatrix} && \xrightarrow{\quad} (X \ Y) + (-Y \ -X) \\
 R_{3,X,Y} &= (X) + (Y) && \xrightleftharpoons[0.01]{0.01} (X \ Y) \\
 R_{4,X} &= \emptyset && \xrightarrow{0.1} (X)
 \end{aligned}$$

for all $X, Y \in \{A, C, G, T\}$.

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

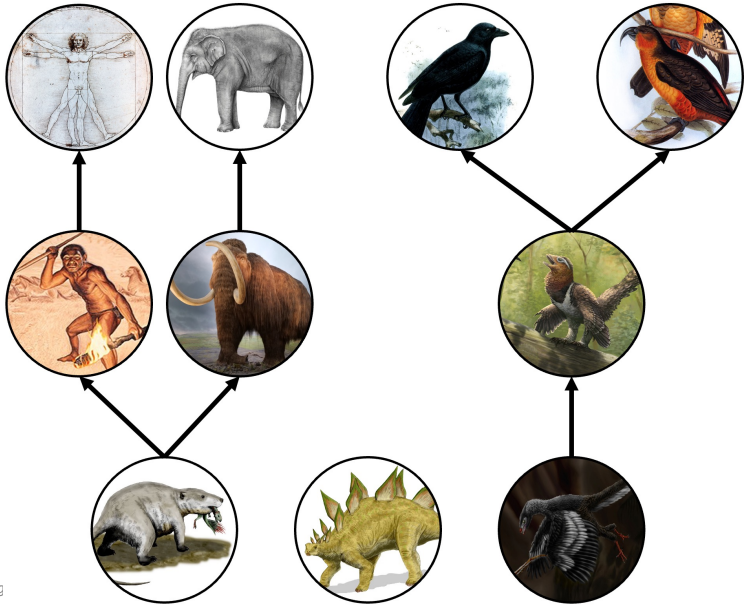
Let $X_0 = \emptyset$.

Natural Evolution



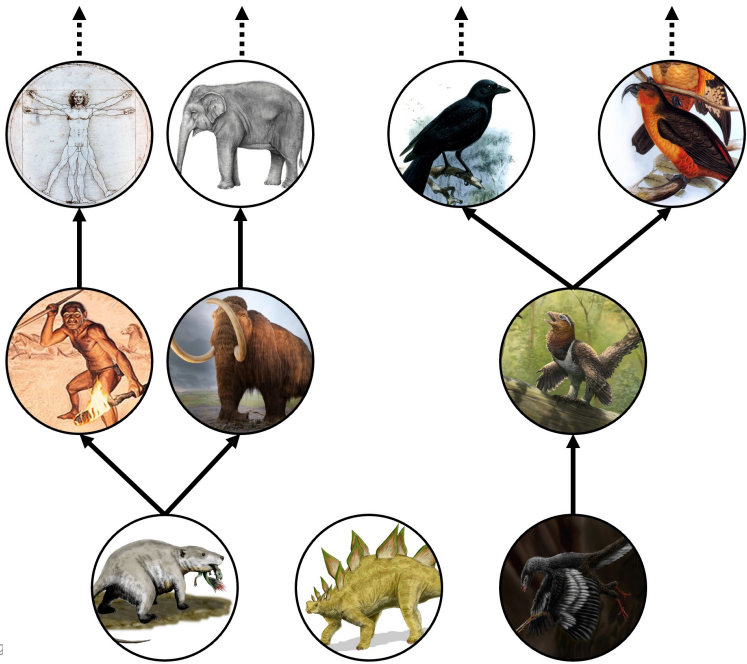
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de.wikipedia.org/wiki/Plesiosaurus#/media/Datei:Plesiosaurus_3DB.jpg
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de.wikipedia.org/wiki/Evolution_der_Säugetiere#/media/Datei:Repenomamus_BW.jpg
de.wikipedia.org/wiki/Mammute#/media/Datei:Victoria_Mammut.P1033804.jpg
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www.australiangeographic.com.au/topics/history-culture/2012/09/elephant-illustrations-the-art-of-science/
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Why does it keep going?



de.wikipedia.org/wiki/Plesiosaurus#/media/Datei:Plesiosaurus_3DB.jpg

de.wikipedia.org/wiki/Stegosaurus#/media/Datei:Stegosaurus_BW.jpg

de.wikipedia.org/wiki/Evolution_der_Säugetiere#/media/Datei:Repenomamus_BW.jpg

de.wikipedia.org/wiki/Mammute#/media/Datei:Victoria.Mammut.P1033804.jpg

st.museum-digital.de/index.php?t=sammlung&gesusa=730

www.australiangeographic.com.au/topics/history-culture/2012/09/elephant-illustrations-the-art-of-science/

de.wikipedia.org/wiki/Vitruvianischer_Mensch#/media/Datei:Vitruvian_Man_by_Leonardo_da_Vinci.jpg

upload.wikimedia.org/wikipedia/commons/6/69/Balaur_bondoc_as_avialan.jpg

indianapublicmedia.org/amomentofscience/crow-thievery-2.php

The Subject of Evolution

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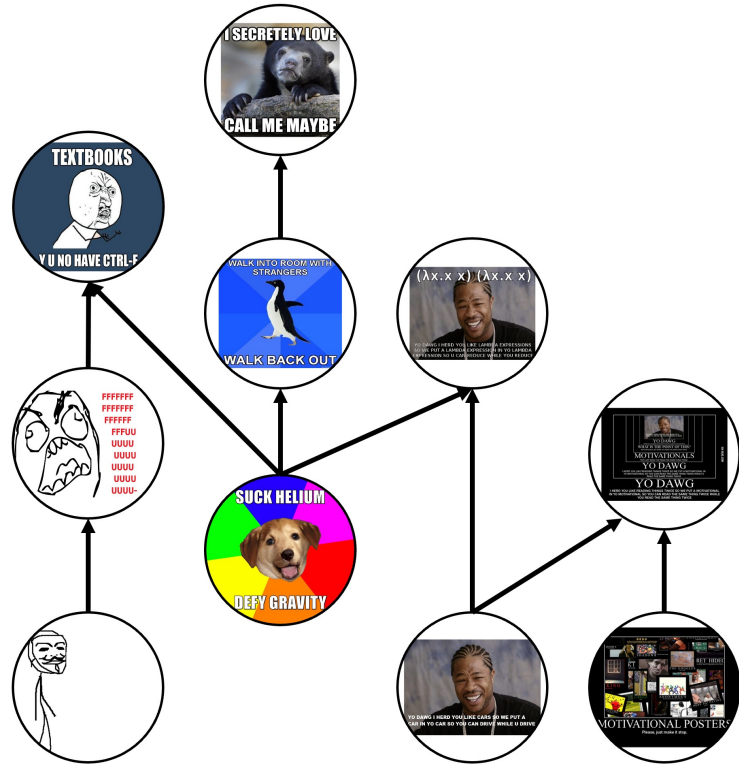
The Gene

in the evolutionary biology sense

Other
(Non-Biological)
Instances of Evolution?

Detour: The Meme

Detour: The (Internet) Meme



source: internet
also pt.slideshare.net/TjerkW/the-rise-of-functional-programming

Dawkins.
The Selfish Gene. (40th Anniversary Edition)
Oxford University Press, 2016.

ebookcentral.proquest.com/lib/ub-lmu/detail.action?docID=4545419

